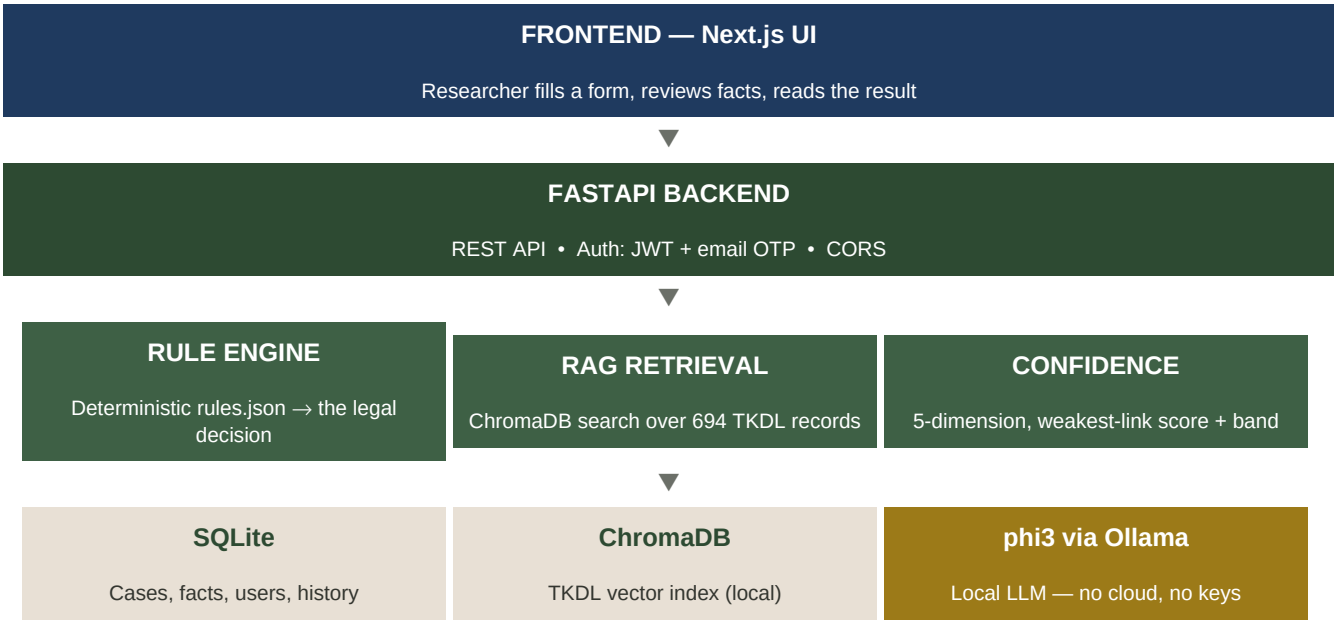
IP-SAKTI Sahayak — Backend Explained

How the engine decides, in plain language | a 1–2 minute read for judges, users and reviewers

Think of IP-SAKTI Sahayak as a friendly first check for anyone making an Ayurvedic product. Before spending money on lawyers, you can find out in minutes whether your formulation is something genuinely new worth protecting, or a well-known traditional recipe — and what approvals it may need. Best of all, it shows its reasoning at every step, in language you can actually understand.

In one line: A backend that tells an Ayurveda startup whether a herbal formulation is **patentable**, is a **classical/known** preparation, or needs **biodiversity (ABS) approval** — and backs every answer with a real rule, a cited traditional-knowledge record, and a confidence score. **The AI never makes the legal call.**

Architecture — the moving parts



The key idea: the AI (phi3) only **reads** the description into structured facts and **explains** the result in plain words. The **classification comes from a deterministic rule engine** — so every answer is auditable and repeatable, not a guess.

Data flow — what happens on one case

1 Create case

User submits a formulation (ingredients, use, claims) → saved to SQLite.

2 Extract facts

phi3 reads the text into 5 yes/no facts (biological resource? commercial? modified?…).

3 Verify facts

User confirms or edits the facts — the cheapest guard against a bad AI read.

4 Rule engine

Confirmed facts + as-of date → deterministic classification + the rule that fired.

5 Retrieve evidence

Hybrid BM25 + vector search finds the most relevant TKDL prior-art records.

6 Validate + score

Check evidence relevance, then a 5-part confidence score → HIGH/MEDIUM/LOW/INSUFFICIENT.

7 Explain

Assemble the full trace: facts → rule → citation → evidence → confidence, for the UI.

→ RESULT, or ABSTAIN / escalate to a human when evidence is insufficient.

The analysis pipeline — under the hood

/cases/{id}/analyze runs a single deterministic pass. Each stage has a clear input and output, is independently unit-tested, and writes its result into the case's stored JSON.

STAGE	INPUT	WHAT HAPPENS

1 · Fact extraction	Case description	phi3 (JSON mode) → 5 boolean facts + source spans + confidence; deterministic keyword fallback if the model is unavailable.
2 · Normalization	Raw facts	Canonicalizes ingredient names via an alias map (e.g. Ashwagandha → Withania somnifera).
3 · Rule engine	Confirmed facts + date	Matches facts against rules.json conditions; returns classification + fired rule(s). Plain Python, no AI.
4 · Evidence retrieval	Case text	Hybrid BM25 + vector search over the TKDL corpus in ChromaDB; returns ranked prior-art records.
5 · Citation validation	Retrieved records	Keeps only records whose retrieval relevance clears a calibrated threshold (≥ 0.3).
6 · Confidence	All upstream scores	5-dimension, weakest-link (min-gated) score → HIGH / MEDIUM / LOW / INSUFFICIENT band.
7 · Explainability	Full analysis	Assembles the trace (facts → rule → evidence → confidence). No new computation — presentation only.

Rule engine (the decision-maker). Rules live in `rules.json` as declarative objects: an `id`, a list of `conditions` (`fact / op / value`), a `conclusion`, a **priority**, a **citation**, and optional `effective_from / effective_to` dates. The engine filters rules by the case's **as-of date** (temporal reasoning), evaluates conditions deterministically, sorts fired rules by priority, and if two equal-priority rules disagree it returns `conflicting_rules_require_review` instead of guessing. The LLM is never allowed to write the classification.

Confidence model. Five dimensions are scored 0–1: **fact** (extraction quality), **rule-fit** (did a rule fire), **evidence** (retrieval strength), **citation** (validation), and the **overall** = `min(the four)` — a **weakest-link** gate, so one weak link caps the whole score. Bands: **HIGH** ≥ 0.80, **MEDIUM** ≥ 0.50, **LOW** ≥ 0.25, else **INSUFFICIENT** → the system abstains and offers human escalation rather than over-claiming.

Retrieval & RAG — grounding every answer

The corpus is **694 real TKDL formulation records** (Ayurveda / Unani / Siddha), embedded once with **sentence-transformers all-MiniLM-L6-v2** (384-dim) and stored in **ChromaDB**. A query is embedded and combined with a **BM25** keyword score into a single hybrid ranking, so results are strong on both meaning and exact terms. The retrieval layer was evaluated offline on **1,540 source-grounded queries**, reaching **MRR ≈ 0.82** and **Hit@5 ≈ 0.88**. Retrieval supplies evidence only — it never decides the outcome, and the same index powers the grounded **Sahayak AI** chatbot.

Data model (SQLite)

TABLE	PURPOSE
FormulationSubmission	The case: submission_id, title, description, jurisdiction, as-of date, tool, status, and the full result_json (classification, confidence, evidence, history, versions).
ExtractedFact	One row per fact: field, value, normalized_value, status (EXTRACTED / CONFIRMED), confidence, source_span.
User	Account: email, bcrypt-hashed password, verification flag, profile and default jurisdiction.
OTPRecord	Time-limited email one-time-passwords for verification / OTP login.
TKDLRecord	Schema for structured TKDL entries used alongside the vector corpus.

Engineering highlights

- **Decision / AI separation** — the legal conclusion comes only from the deterministic rule engine; the LLM is structurally prevented from writing it.
- **Fully local & private** — phi3 runs through Ollama's native API with a capped context window (num_ctx) so it fits modest RAM; no cloud calls, no API keys, JSON-mode output for reliable parsing.

• **Human-in-the-loop** — users confirm every extracted fact before analysis, and low-confidence cases abstain and route to a human reviewer.

• **Grounded, honest chatbot** — intent routing replies briefly to greetings, redirects off-topic questions, and answers Ayurveda/IP questions from retrieved TKDL evidence.

• **Auditable & versioned** — every case keeps a full history and re-analysis versions; each result carries its trace and citations.

• **Tested** — unit + integration test suite covers the rule engine, confidence, normalization, temporal logic and citation validation.

Main API endpoints

METHOD	ENDPOINT	WHAT IT DOES
POST	`/auth/signup · /verify-otp · /login`	Register, verify email OTP, sign in (JWT).
POST	`/cases`	Create a research case.
GET	`/cases · /cases/{id}`	List all cases / open one case.
POST	`/cases/{id}/facts`	Extract facts (AI) or save the user-confirmed facts.
POST	`/cases/{id}/analyze`	Run the pipeline → classification, evidence, confidence.
GET	`/cases/{id}/explain`	The step-by-step reasoning trace.
GET	`/cases/{id}/report`	Full exportable case report.
POST	`/cases/{id}/escalate`	Flag the case for a human reviewer.
POST	`/assistant/chat`	Grounded Q&A; chatbot (phi3 + TKDL evidence).

The backend in 60 seconds (how it works): It's a **FastAPI** service using **SQLite** for cases and **ChromaDB** for search. Analyzing a case runs a **7-stage pipeline**. (1) The local **phi3** model (via Ollama) reads the description into five yes/no facts. (2) The user **confirms** them, so a bad AI read never reaches the decision. (3) The **deterministic rule engine** matches those facts against rules.json (with as-of-date filtering) and returns the classification and the exact rule that fired — the **only** place the legal call is made. (4) **Hybrid retrieval** (BM25 + sentence-transformer vectors over 694 TKDL records) pulls the closest prior art, then (5) each citation's relevance is validated. (6) A **5-dimension, weakest-link confidence** score sets a band; if it's insufficient the system **abstains and offers human escalation**. (7) The full trace is assembled for the UI. Every step is deterministic and auditable — the AI only reads and explains, it never decides.

Say it in 60 seconds: "A user describes their Ayurvedic product. Our local AI turns that into structured facts, which the user confirms. A **deterministic rule engine** — not the AI — then decides the IP status and shows exactly which rule and law applied. We back it with **real TKDL prior-art evidence** retrieved from a 694-record search index, score our **confidence**, and if the evidence is weak we **abstain and escalate to a human**. Everything runs offline on a local model — no cloud, no API keys, fully auditable."

Stack: FastAPI · SQLite · ChromaDB · sentence-transformers (all-MiniLM-L6-v2) · rank-bm25 · phi3/Ollama. This is a prototype for research support — not legal advice.